# Siemens S700K Point Machine
# Motor Current Draw Diagnostics

Normal Operating Parameters, Fault Indicators, and Escalation Criteria

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| **Document ID:** | MAN-PM-002 |
| **Asset Type:** | Point Machine |
| **Language:** | EN |
| **Revision:** | Rev B \| 2024-09-01 |
| **Status:** | APPROVED FOR USE |

## 1. Overview

Motor current draw is a primary health indicator for the S700K point machine. Changes in current consumption reflect the mechanical load on the drive gearbox and can provide early warning of bearing wear, lubrication failure, debris ingestion, or ice build-up in the switch bed. This document defines normal operating parameters and the criteria for scheduling preventive maintenance based on current trending.

*[Schematic: S700K motor current waveform during a normal switch operation]*

*Figure: S700K motor current waveform during a normal switch operation*

## 2. Normal Operating Parameters

Under normal operating conditions at ambient temperatures between -10 C and +40 C, the S700K drive motor exhibits the following current profile during a full switch operation:

| Phase | Current | Duration | Status |
|---|---|---|---|
| Inrush (start) | Up to 4.5 A peak | First 150 ms | Normal - motor acceleration |
| Steady-state operation | 2.1 - 2.8 A | Remainder of throw | Normal |
| Locked position (holding) | 0.4 - 0.8 A | Continuous | Normal |
| Sustained > 3.5 A | > 3.5 A | > 500 ms | FAULT - excessive resistance |
| Sustained > 4.5 A | > 4.5 A | Any duration | FAULT - immediate inspection |

**WARNING: Sustained motor current above 3.5 A for more than 500 ms during normal operations indicates excessive mechanical resistance. Log the fault and schedule inspection within 7 days. Do not wait for the current to trip the overload protection.**

## 3. Trending Analysis and Preventive Maintenance Trigger

Single-event high current readings may result from transient causes such as debris in the switch bed or low-temperature grease hardening. However, a consistent upward trend in steady-state current across multiple switch operations is a reliable indicator of progressive gearbox bearing wear. The CMMS telemetry dashboard plots daily mean current and flags the following thresholds:

| Trending Condition | Threshold | Recommended Action |
|---|---|---|
| Week-on-week increase | > 15% per week | Schedule gearbox inspection within 7 days (Sec 6.2) |
| 30-day cumulative increase | > 40% | Expedite gearbox lubrication and bearing check |
| 30-day cumulative increase | > 60% | Treat as high-risk; bearing replacement likely required |
| Single spike (transient) | > 3.5 A, non-recurring | Check switch bed for debris or ice; monitor |

Section 6.2 of this manual details the gearbox lubrication and bearing inspection procedure. Spare parts required: Gearbox bearing kit P/N S700K-BRG-04 and one Siemens LF-200 grease cartridge. Estimated repair time: 90 minutes.

*NOTE: Low ambient temperatures (below -5 C) can cause LF-200 grease to harden, resulting in elevated current even on healthy bearings. Cross-reference ambient temperature logs before scheduling maintenance for* *cold-weather current spikes. See MAN-PM-004 for winter lubrication procedures.*